

Krydo

Prove you qualify — without revealing what you have

Verifiable credentials ■ Zero-knowledge proofs ■ Stellar Soroban

Live demo: <https://krydo-stellar.vercel.app>

GitHub: <https://github.com/nishant-uxs/krydo-stellar>

The Problem

Verification today over-collects

A lender asking “*Do you earn at least INR 10 L?*” often receives:

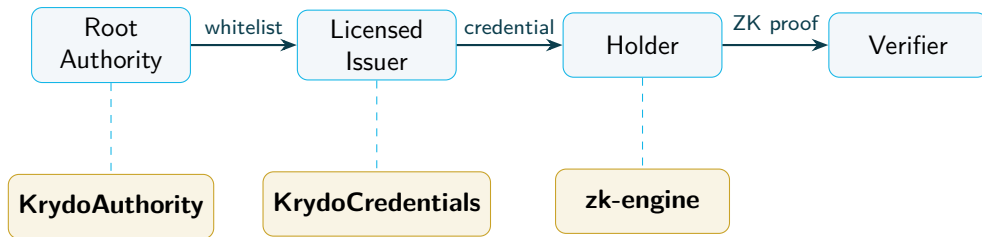
- Exact salary & employer
- Bank statements
- PAN / national ID copies
- Data cannot be revoked once leaked
- Re-sold across brokers
- High compliance & breach risk

What we need

Prove a **predicate** ($\text{score} \geq 700$, $\text{income} \geq 10\text{L}$) — not the underlying value.

Our Solution

Krydo is a three-tier trust network on **Stellar Soroban**:



End-to-End Flow

- 1 **Issuer** (whitelisted on-chain) signs `issue_credential` → hash on ledger, plaintext off-chain
- 2 **Holder** generates a **sigma-protocol ZK proof** in the browser (Pedersen + Fiat-Shamir)
- 3 **Verifier** calls `POST /api/zk/verify` — learns only whether the predicate holds
- 4 Optional: holder wallet-signs `KrydoAudit.anchor` for a public audit trail

Design principle

Blockchain = source of truth for *what exists*. Firestore = fast query layer. Private keys never leave the wallet.

Zero-Knowledge Layer

Not SNARKs — real sigma protocols

- Pedersen commitments $C = vG + rH$
- Fiat-Shamir non-interactivity
- No trusted setup, no circuits
- Proofs in milliseconds (browser + API)

Proof types: range_above, range_below, equality, membership, selective_disclosure

Example

Credential: credit score **782**

Proof shared: **score** $\geq$ **750**

Verifier learns: predicate is **true**

Verifier does **not** learn: 782

Soroban Smart Contracts

Contract	Responsibility
KrydoAuthority	Root-controlled issuer whitelist (add_issuer, revoke_issuer)
KrydoCredentials	Credential hashes, issuance, revocation on-chain
KrydoAudit	Stateless event log for wallet-signed anchors (requests, roles, ZK)

- Auth: **Sign-in-with-Stellar** (SEP-53) → JWT
- Every state-changing tx: in-app confirm → **wallet signature**
- Wallets: Freightier, xBull, Lobstr, Hana via Stellar Wallets Kit

Deployed on Stellar Testnet

Network: Testnet **Deployed:** 11 July 2026 **Explorer:**

<https://stellar.expert/explorer/testnet>

Role / Contract	Address (StrKey)
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Root authority	GC5VBHY5DWV7NTL4PCQL3XG0E4FY2DJHM2JYLRC6YS2IHYTPDZ4D0FIU
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KrydoAuthority	CDBYAPYW340CTNMTT3H75XLC2JRA7HD4ULA5YVJ75SISM67GPCD6FNK2
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KrydoCredentials	CD7PFDNVL2V4S04WQWP225CJRYV24GIGM3ZALS2WHIEM6UINQK7KJCOV
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KrydoAudit	CALAZRCMK2POSKYEXA7LB3SPIBU5555SD5LX53LHC2YYCSZMYEQVXJSX
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RPC: <https://soroban-testnet.stellar.org>

Technology Stack

On-chain

- Soroban (Rust / WASM)
- Stellar Testnet
- Sub-cent tx fees

Backend

- Node.js + Express
- Firestore (mirror + queries)
- Zod validation, JWT auth

Frontend

- React + Vite + TypeScript
- Stellar Wallets Kit
- Client-side ZK engine

Quality

- 168 unit tests, CI on GitHub Actions
- Open source (MIT)

Security & Trust Model

- Four roles: Root, Issuer, Holder, Verifier
- No role can elevate itself
- On-chain `require_auth` for issuance
- Claim summary vs. value consistency checks
- Issuer list synced from KrydoAuthority

MVP status

Testnet deployment. External crypto audit and multi-sig root recommended before production finance use.

Roadmap

Phase	Milestone
Done	Testnet contracts, multi-wallet SIWS, wallet-signed txs, live demo
Next	Mainnet deploy (~100–102 XLM operating budget), demo issuers, waitlist
Future	Multi-sig root, encrypted credential store (IPFS/Arweave), mainnet partners

Mainnet operating budget (approx.)

100–102 XLM total — contract deploy & integration testing ~2–5 XLM; balance held as reserve for demo issuers, storage rent (TTL), and 12-month on-chain operations.

Try it now

<https://krydo-stellar.vercel.app> Connect wallet on **Testnet**

Thank you

Questions?

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Repo: <https://github.com/nishant-uxs/krydo-stellar>

Docs: [DOCUMENTATION.md](#)